

This exam paper is part of a set used to assess student abilities as follows:

		Q1- Q8
CLO 1	Demonstrate comprehensive understanding of complex numbers, matrices, sequences, series, and vectors, including their meanings and underlying principles.	✓
CLO 2	Apply the acquired knowledge and understanding of mathematics to solve a diverse range of problems effectively and appropriately.	✓
CLO 3	Proficiently calculate, analyze, and employ fundamental mathematical principles in the context of scientific disciplines..	✓
CLO 4	Cultivate moral and ethical values, respect for others' rights, actively listen to diverse opinions, and develop effective teamwork skills.	✓
CLO 5	Practice and enhance the computational skills through the proficient use of information technology tools.	✓

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1. Determine the general term (a_n) of the following sequences

1.1 The sequence 1, 6, 13, 22, 33, ... (3 points)

1.2 The sequence $\frac{2 \cdot 1}{1}, \frac{3 \cdot 4}{4}, \frac{4 \cdot 9}{7}, \frac{5 \cdot 16}{10}, \dots$ (3 points)

1.3 The sequence $\frac{1}{1}, \frac{3}{5}, \frac{6}{12}, \frac{10}{22}, \frac{15}{35}, \dots$ (4 points)

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2. Let a, b, c be real numbers which $\frac{1}{a}, \frac{1}{2b}, \frac{1}{3c}$ is a geometric sequence

and a, b, c is an arithmetic sequence. Find $\frac{a}{c} + \frac{c}{a}$.

(7 points)

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3.1. Find the sum of numbers which are divisible by 4 from 9 to 175.

(5 points)

3.2. Let $\sum_{n=1}^4 (bn^2 - 5) = 250$. Find the sum of first b terms of $2 \cdot 5 + 4 \cdot 6 + 6 \cdot 7 + 8 \cdot 8 + \dots$

(using $\sum$ notation)

(5 points)

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4.1. Given $a_n = (-1)^n (\sqrt{n^2 + 4n} - \sqrt{n^2 - 2})$.

Find $\lim_{n \rightarrow \infty} a_n$ or state that it does not exist and explain your reasoning.

(4 points)

4.2. Given $a_n = \frac{\sqrt{8n} - \sqrt[3]{n+2} + 7}{1 - 2\sqrt[4]{n} - \sqrt{2n}} + \frac{5^{n+1}}{3^n - 5^n}$.

Find $\lim_{n \rightarrow \infty} a_n$ or state that it does not exist and explain your reasoning.

(4 points)

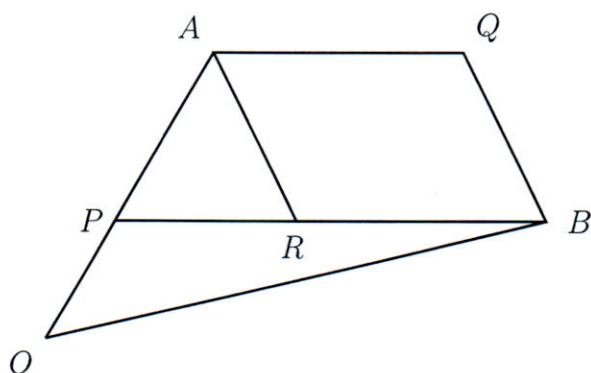
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4.3. If c is a constant and $\lim_{n \rightarrow \infty} \frac{c^3 n - 3n^2 + 6cn^3}{(n+1)^3} = 1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$. Determine the value of c .

(6 points)

5. From the figure, let $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$ such that $\overrightarrow{OP} : \overrightarrow{PA} = 1 : 2$ and $\overrightarrow{PR} : \overrightarrow{RB} = 2 : 3$.

If $ARBQ$ is the parallelogram, find $\overrightarrow{RB}$ and $\overrightarrow{OQ}$ in the forms of $\vec{a}$ and $\vec{b}$. (7 points)



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6. Assume that $\vec{u} = 2\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{v} = -\hat{i} + \hat{j} + 3\hat{k}$

6.1 Find the vector of length 2 which has the opposite direction as $\vec{u}$.

(4 points)

6.2 Find the vector of length $|\vec{u} \cdot \vec{v}|$ which has the same direction as $\vec{u} - 2\vec{v}$.

(6 points)

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7. Assume that $\vec{u}$, $\vec{v}$, and $\vec{w}$ are the vectors such that $\vec{u} + \vec{v} + \vec{w} = \vec{0}$ and $|\vec{u}| = 1$, $|\vec{v}| = 4$, $|\vec{w}| = 5$.

Find $3|\vec{u} - \vec{v}| + |\vec{v} - \vec{w}|$.

(8 points)

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8. Given $\vec{u} = 3\hat{i} - \hat{j} + 2\hat{k}$, $\vec{v} = 2\hat{i} - 4\hat{j} + 3\hat{k}$, and $\vec{w} = -2\hat{i} - (2 - 3\lambda)\hat{j} + \lambda\hat{k}$.

8.1. Compute $\vec{u} \cdot \vec{v}$ and $\vec{u} \times \vec{v}$.

(5 points)

8.2. Find the value of λ such that $\vec{u}$ and $\vec{w}$ are orthogonal.

(4 points)

8.3. Find the value of λ such that $\vec{w} \cdot (\vec{u} \times \vec{v}) = 0$.

(5 points)

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